ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

INTRODUCTION TO INFORMATION SYSTEMS

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) Information Technology and Computer Science, Level 2. Section A: objective questions (30) · Section B: computer concepts, components of an information system, software and file management, the Internet, ethics and health (70).

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INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 Raw, unprocessed facts are called …

A information. B knowledge. C output. D data.

1.1.2 Which ONE of the following is an input device?

A Printer B Scanner C Monitor D Speaker

1.1.3 The part of the computer that carries out instructions and calculations is the …

A RAM. B hard drive. C power supply. D CPU.

1.1.4 Which type of memory loses its contents when the computer is switched off?

A ROM B RAM C Flash memory D SSD

1.1.5 1 gigabyte (GB) is approximately equal to …

A 1 024 bytes. B 1 024 KB. C 1 024 MB. D 1 024 TB.

1.1.6 Windows, Android and Linux are examples of …

A operating systems. B application software.
C utility software. D programming languages.

1.1.7 An information system that records the day-to-day sales of a shop is a …

A decision support system (DSS). B transaction processing system (TPS).
C expert system. D management information system (MIS).

1.1.8 A program that pretends to be useful but secretly harms the computer is a …

A worm. B firewall. C cookie. D Trojan horse.

1.1.9 The file extension .xlsx indicates a …

A spreadsheet. B word-processing document.
C presentation. D image.

1.1.10 The address of a web page, such as https://www.gov.za, is called a …

A ISP. B IP. C URL. D LAN.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 Software is the physical part of a computer that you can touch.

1.2.2 A procedure is one of the five components of an information system.

1.2.3 A byte consists of 8 bits.

1.2.4 A solid-state drive (SSD) has moving parts.

1.2.5 Information must be accurate, relevant and timely to be useful.

1.2.6 A firewall can help to stop unauthorised access to a network.

1.2.7 Copying licensed software for friends without permission is legal.

1.2.8 Cloud storage saves files on remote servers that you access over the Internet.

1.2.9 Ergonomics is the study of how computer viruses spread.

1.2.10 A folder can contain other folders (sub-folders).

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Software that is free to use and whose code may be changed	A	repetitive strain injury (RSI)
1.3.2	Main circuit board that connects all the components	B	information
1.3.3	A copy of files kept in case the originals are lost	C	backup
1.3.4	Fraudulent e-mail that tries to trick you into giving your password	D	LAN
1.3.5	Company that provides access to the Internet	E	ISP
1.3.6	Network covering a small area such as one building	F	motherboard
1.3.7	Processed data that has meaning	G	browser
1.3.8	Software that helps to maintain the computer, e.g. a disk cleaner	H	phishing
1.3.9	Program used to view web pages	I	open-source software
1.3.10	Pain in the wrist caused by repeated typing	J	utility program

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: COMPUTER CONCEPTS AND HARDWARE

2.1 Explain the difference between *data* and *information*. Give an example of each from a school environment. (4)

2.2 Draw a labelled block diagram of the information processing cycle (input, processing, output and storage). (4)

2.3 Give TWO differences between RAM and ROM. (4)

2.4 A learner wants to buy a computer for programming and video editing. Name THREE hardware specifications the learner should look at and explain why each one is important. (6)

2.5 Convert the following: (a) 3 KB to bytes (b) 2 048 MB to GB (2)

[20]

QUESTION 3: INFORMATION SYSTEMS

3.1 Name the FIVE components of an information system. (5)

3.2 A clinic uses a computerised system to register patients and keep their medical records. Explain how EACH of the five components in QUESTION 3.1 applies to this clinic. (5)

3.3 Distinguish between a transaction processing system (TPS) and a management information system (MIS). (4)

3.4 Give THREE characteristics of good-quality information. (3)

3.5 State THREE advantages of using a computerised information system rather than a paper-based system. (3)

[20]

QUESTION 4: SOFTWARE AND FILE MANAGEMENT

4.1 Distinguish between system software and application software. Give ONE example of each. (4)

4.2 Name THREE functions of an operating system. (3)

4.3 Study the file path below and answer the questions.

C:\Users\Thandi\Documents\NCV\Level2\Assignment1.docx

4.3.1 What is the name of the file? (1)

4.3.2 On which drive is the file stored? (1)

4.3.3 In which folder is the file stored directly? (1)

4.3.4 Which application would normally open this file? (1)

4.4 Explain why it is important to give files and folders meaningful names. (2)

4.5 Explain the difference between *Save* and *Save As*. (2)

[15]

QUESTION 5: THE INTERNET, SECURITY, ETHICS AND HEALTH

5.1 Name TWO ways in which a student can connect a computer to the Internet. (2)

5.2 Give FOUR guidelines (netiquette) for writing a professional e-mail. (4)

5.3 Suggest THREE ways to protect a computer against viruses and other malware. (3)

5.4 Explain what *plagiarism* is and how a student can avoid it when using information from the Internet. (2)

5.5 Name TWO health risks of working at a computer for long periods and give ONE way to prevent each. (4)

[15]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER